

Long-term Use of β -Carotene, Retinol, Lycopene, and Lutein Supplements and Lung Cancer Risk: Results From the VITamins And Lifestyle (VITAL) Study

High-dose β -carotene supplementation in high-risk persons has been linked to increased lung cancer risk in clinical trials; whether effects are similar in the general population is unclear. The authors examined associations of supplemental β -carotene, retinol, vitamin A, lutein, and lycopene with lung cancer risk among participants, aged 50–76 years, in the VITamins And Lifestyle (VITAL) cohort Study in Washington State. In 2000–2002, eligible persons ($n = 77,126$) completed a 24-page baseline questionnaire, including detailed questions about supplement use (duration, frequency, dose) during the previous 10 years from multivitamins and individual supplements/mixtures. Incident lung cancers ($n = 521$) through December 2005 were identified by linkage to the Surveillance, Epidemiology, and End Results cancer registry. Longer duration of use of individual β -carotene, retinol, and lutein supplements (but not total 10-year average dose) was associated with statistically significantly elevated risk of total lung cancer and histologic cell types; for example, hazard ratio = 2.02, 95% confidence interval: 1.28, 3.17 for individual supplemental lutein with total lung cancer and hazard ratio = 3.22, 95% confidence interval: 1.29, 8.07 for individual β -carotene with small-cell lung cancer for >4 years versus no use. There was little evidence for effect modification by gender or smoking status. Long-term use of individual β -carotene, retinol, and lutein supplements should not be recommended for lung cancer prevention, particularly among smokers.